Timothy Hsu

timmyhsu07@gmail.com | (862) 324-6815 | linkedin.com/in/timhsu7 | github.com/timmyhsu07 | timmyhsu07.github.io | U.S. Citizen

EDUCATION

Rutgers University — Honors College New Brunswick, NJ
B.S. in Computer Science and Mathematics, ***GPA 3.94/4.00*** *Sep. 2024 – May 2028*

- Coursework: Data Structures and Algorithms, Discrete Structures, Computer Architecture, Probability and Statistics
- Activities: Break Through Tech AI, Quantitative Finance Club, Rutgers Chinese Student Organization

EXPERIENCE

Machine Learning Researcher May 2026 – Aug. 2026
TRACE AI Lab, Rutgers University (Aresty Summer Science Program) *New Brunswick, NJ*

- Built **GLIMPSE**, an interactive Python tool that maps every equally-accurate model for a dataset onto one 2D view; **deployed as a live web app** on Vercel with serverless endpoints and precomputed model fits.
- Designed **ECLIPSE**, a projection with an exact zero-error inverse, reaching **0.99+ decision fidelity** (vs. 0.88 for PCA) and surfacing **39% of model disagreement** vs. 0.7% for PCA across 7 datasets.
- Benchmarked 4 dimensionality-reduction methods over 5,000+ records and structured the codebase into 4 modular libraries behind shared interfaces, cutting onboarding time for a new dataset or algorithm by **roughly 80%**.

Lead Programmer and Research Assistant Sep. 2025 – Present
Eleanna Kara Neurodegenerative Disease Lab, Rutgers University *Piscataway, NJ*

- Built Python image-analysis pipelines over 800+ microscopy images, **recovering 233 cells** missed by manual counting in a 10-image audit.
- Implemented segmentation, thresholding, and contour detection in OpenCV and scikit-image to extract 15–20 cell features (soma area, circularity, neurite length), raising analysis throughput **5x**.
- Automated quality control, visualization, and summary statistics across 8+ experimental batches, making downstream analyses reproducible.

PROJECTS

gamma-exit — Options Research Backtesting Engine Apr. 2026 – Jul. 2026
Tests Ramkumar (2025), "Gamma Scalping–Theta Decay Trade-Off" | Python, NumPy, SciPy, QuantLib, pytest | ***GitHub***

- Built a simulation engine that prices and tracks hedged option positions day by day, validating its math against the **QuantLib** reference library to **1e-8** across 3,000+ simulated runs.
- **Found an error in the paper's derivation** by writing an invariance test that varied one parameter in isolation; P&L stayed flat where the paper required it to move, isolating a drift term that self-financing accounting should have cancelled.
- Designed the API so future data cannot leak into a simulation: each strategy receives an immutable per-day snapshot, and the one component with lookahead lives in a separate module behind an incompatible interface.
- Shipped with a **172-test suite in CI**, config-driven runs, and a write-once cache for reproducible results; a causal strategy recovered **72% of the perfect-foresight ceiling** (90% CI: 66–77%) in regime-shift scenarios.

MAGI — Antimicrobial Resistance Decision Support Jul. 2026
HackNation 2026 | Python, scikit-learn, Streamlit, Parquet | ***GitHub***

- Built a **3-module pipeline** converting assembled bacterial genomes into AMRFinderPlus features, calibrated per-drug predictions, and evidence-linked reports across 3 antibiotics.
- Implemented grouped cross-validation that keeps related samples from splitting across train and test, probability calibration, and an out-of-distribution check so **low-confidence cases abstain instead of guessing**.
- Shipped a Streamlit decision app with swappable demo and live backends plus gene-level evidence traces, validated by a **109-test suite** covering calibration, gating, and abstention paths.

Courtsense — Live Basketball Trading Model Sep. 2025
Quant Challenge 2025, ***Top 2% Finalist*** *| Python, scikit-learn, pandas |* ***GitHub***

- Trained a 200-tree random forest over 10,000+ observations and 14 engineered features, improving predictive accuracy **25%+ over baseline** models.
- Sized positions with the Kelly Criterion over 1,000+ in-game events per match, reaching **18% simulated ROI** in backtests across varying game states and volatility conditions.

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, JavaScript, HTML/CSS
Machine Learning & Data: PyTorch, scikit-learn, NumPy, pandas, OpenCV, scikit-image, autoencoders, dimensionality reduction (PCA, t-SNE, UMAP), model calibration, cross-validation
Tools: Git, GitHub Actions, Docker, pytest, Streamlit, Jupyter, Matplotlib, LaTeX, DuckDB, Parquet, QuantLib, Vercel
Spoken Languages: English (Native), Chinese (Native)